Participation Costs, Search Time, and Match Quality in the Housing Choice Voucher Program*

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Abstract

Many social safety net programs struggle with incomplete program take-up, oftentimes due to high participation costs. The Housing Choice Voucher Program involves a complex, multi-step take-up process and coordination with the private rental market, creating considerable barriers to entry for participants. Potential voucher recipients also must navigate an increasingly difficult housing search process in a limited amount of time, making it even more challenging to use their vouchers at all, and to find a home that meets their preferences. In this paper, I evaluate the effects of two unique features of the voucher program: the timing of voucher issuance, and variation in how long voucher recipients are given to search. In the aggregate, the timing of voucher issuance has only a mechanical effect on the likelihood of using a voucher within a certain time frame, and no effect on the chance of a participant ever using their voucher. However, housing authorities vary in their time restrictions on voucher use and their leniency in granting extensions. At housing authorities that do not offer extensions, end-of-month issuance results in a sizeable 6 percentage point decline in voucher success rates. Households at these PHAs see shorter search times, and are more likely to end up in over-crowded housing situations.

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1 Introduction

Social programs often struggle with incomplete take-up. In many cases, this can come from difficulties in learning about programs, documenting eligibility, and meeting all necessary requirements. However, the Housing Choice Voucher program, the largest federal rental assistance program in the country, involves additional challenges for participants. In the voucher program, participants must find and rent eligible homes on the private rental market in order to receive their subsidies. This means that the already challenging private market housing search process is inserted into the program take-up equation. In recent years, the tightening of rental markets across the country has intensified the difficulty of using vouchers; in 2022, only 57% of households who were issued vouchers were able to use them at all.

Even for households who are successful in using their vouchers, there is evidence of growing search costs and constrained options in the housing market. Search times, or the amount of time between voucher issuance and leasing a home with that voucher, have grown sharply as voucher recipients contend with rapidly rising rents, a shortage of affordable housing units, and the difficulty of finding landlords who will participate in the program. Growing search times are troubling, as in general voucher recipients have only a set period of time in which they must use their voucher, or the voucher will be returned to the housing authority. And despite the program's core goal of opening up more neighborhoods to low-income households, most voucher recipients tend to live in similar neighborhoods to unassisted low-income households (Galvez, 2011; Wood et al., 2008).

In this paper, I study the effects of two unique elements of housing voucher take-up. The first is the timing of voucher issuance. The exact timing of voucher issuance is random, and while vouchers can be issued on any day on the month, households are most likely to lease up on the first of the month. Since voucher take-up requires interaction with the private housing market, the day of the month that a voucher is issued can have mechanical effects on voucher take-up, where households who are issued a voucher at the beginning of the month may be more likely to use their voucher within the time frame that they are given to search. The second unique feature of voucher take-up is that the public housing authorities (PHAs) that administer the program vary in how long they give households to search, and how flexible they are in offering extensions. At PHAs with stricter search time policies, this has the potential to interact with issuance timing and affect the ability of households to use their vouchers before they expire. There also may be implications for the quality of the match that households receive, given that they may have fewer effective days to search if their issuance falls at the end of the month.

I find that new voucher recipients who are issued a voucher at the end of the month are less likely to use their voucher in 60, 90, or 120 days. These effects have persisted over time, and as search times have grown, the effects of end-of-month issuance are observed at even longer search windows. However, there is no effect on the ability of households to ever use their voucher. I also find that, in the aggregate, there are no observable effects on housing match quality for those who are successful in using their vouchers, measured as the ability to reach lower poverty neighborhoods, the length of time that a household remains in a housing unit, and the relative size of the unit that a household leases.

There are, however, important differences in the effects of end-of-month issuance with regard to PHA search time policies. Housing authorities that administer the voucher program are required to give households at least 60 days to search, but in practice, many PHAs allow for much longer, through longer initial search times or extension policies. To measure how the effects of issuance timing may vary by PHA search time policies, I collect original data from PHA administrative plans on search time policies, and use these policies to test for heterogeneity in effects. I find that end-of-month issuance leads to a sizable 6 percentage point decline in the probability of a household ever using their voucher at PHAs with 60 day initial search terms that do not offer extensions, the strictest category of search time policies.

There is also evidence of differential effects on match quality. While PHAs with longer search terms or more generous extension policies, there is some evidence of positive effects on match quality, while PHAs with stricter search time policies see negative effects. At PHAs that do not offer any type of extensions, there is suggestive evidence that households may find it more difficult to move to a broader set of neighborhoods, as households are more likely to move to a higher poverty neighborhood than where they lived prior to receiving their voucher. Additionally, successful households are more likely to lease up into smaller units than they qualify for, compared to households who receive their voucher earlier in the month.

This paper contributes to several strands of literature, and carries important policy implications. First, I leverage two novel data sources. The first is confidential administrative data on voucher success rates and search times. This data source has hugely increased the ability to measure these key program metrics, and allows for measuring success rates for close to 1,000 PHAs, a considerably larger sample than past studies. Second, I collect original data on PHA search policies from administrative plans, creating the first source of consolidated information on how long voucher recipients are given to search. This is particularly important as increasing search time may be a policy lever that both HUD and PHAs use to boost success rates, given the sharp drop in success rates and increases in search costs in recent years.

Additionally, this paper contributes to the existing literature on administrative burden and barriers to program participation. Most closely related is Homonoff and Somerville (2021), which documents how the timing of re-certification interview can affect SNAP participation. More broadly, an existing strain of literature documents how high participation costs can effect program take-up (Currie, 2006; Chetty et al., 2013; Bettinger et al., 2012; Deshpande and Li, 2019). This is the first paper to adopt this framework to housing assistance, enhancing our understanding of participation costs in the context of a program that requires interaction with the housing market and a private third party.

Lastly, I contribute to a growing literature on housing voucher take-up and the housing search process. This paper is one of the first in a small but growing set of paper to estimate voucher success rates (Ellen et al., 2024b; Chyn et al., 2019; Finkel and Buron, 2001). My results on housing match quality add to a body of work that studies how the voucher program and other mobility programs can better meet their goals of encouraging neighborhood choice (Bergman et al., 2024; DeLuca et al., 2023; Ellen et al., 2024a). Lastly, I provide new evidence on how search constraints can affect the ability of low-income households to find housing that meets their needs and preferences (Harvey et al., 2020; Krysan and Crowder, 2017).

The paper proceeds as follows. Section 2 lays out the institutional context, including detailing the housing voucher take-up process. Section 3 describes the administrative data used to calculate success rates and the originally collected data on PHA search policies. Section 4 presents descriptive facts on success rates, search times, and PHA search time policies. Section 5 shows results on the effects of issuance timing on success rates, and section 6 shows the effects of issuance timing on measures of match quality. Section 7 shows heterogeneity in results on success rates and match quality by PHA search time policies, and section 8 discusses the implications of these findings.

2 Institutional context: Housing Choice Voucher program

The Housing Choice Voucher program is the largest federal rental assistance program in the country. In 2022, more than 5 million individuals in 2.3 million households received vouchers. While the program is funded federally by the Department of Housing and Urban Development (HUD), it is administered locally by more then 2,000 public housing authorities (PHAs). PHAs have discretion over the daily operation of the program, but abide by certain federal guidelines set by HUD.

Vouchers can be transformative for low-income households who are able to use them. Once a household enters the voucher program, they generally pay 30 percent of their income towards rent, and the local housing authority covers the rest, up to a certain subsidy limit, called the payment standard. Rent burdens in the voucher program are significantly lower than those of unassisted, low-income renters (Ellen and Torrats-Espinosa, 2020). Vouchers have also been shown to reduce homelessness, reduce over-crowding, and improve children's performance in school (Jacob and Ludwig, 2012; Mills et al., 2006; Wood et al., 2008; Schwartz et al., 2020).

However, the program struggles with two main issues. This first is that many households who receive vouchers are not able to use them. In recent years, as much as 40 percent of households who receive vouchers must return them to the housing authority as they are unable to find and rent a qualified home within the allotted time they are given. Second, the voucher program has not consistently achieved a core goal of allowing voucher holders to live in a diverse set of neighborhoods (Galvez, 2011; Pendall, 2000). Both of these issues can reflect the difficulty that voucher recipients may have during the search process.

2.1 Take up in the voucher program

Take-up in the voucher program is a multi-step process, often spanning multiple years. First, voucher holders must apply to join a waitlist at a PHA. Many PHAs will close their waitlists due to outsized demand for the program, so in order to even apply, participants must often wait for the PHA in their local market to open their wait list (Hembre and Urban, 2023). At this point, households apply, and if they are certified to qualify for the program, they are placed on the wait list. While wait times vary widely across the country, they are generally quite long. On average, households wait 2.5 years, but at some housing authorities included in the sample for this paper, average wait times can be as long as 8 years (Acosta and Gartland, 2021).

Once a household is selected off the waitlist, they are contacted by the PHA. At this point, they confirm their eligibility and interest in the program, and provide documentation. Only then are they issued a search voucher. A search voucher is a promise of receiving the subsidy, given the voucher recipient is able to secure a qualifying unit. At this point, households must be given a minimum of 60 days to search, though many PHAs offer longer search windows. PHAs also have discretion in granting extensions. Many PHAs offer some sort of automatic extensions, and virtually all PHAs grant extensions for a wide variety of acceptable reasons.

Households now must find a unit that meets all programmatic requirements. The unit must have a rent that falls within the payment standard, and must pass a housing quality inspection.[1] The landlord for the unit must also be willing to rent to a voucher holder and participate in the program; once a household leases up with their voucher, the housing authority pays the government portion of the rent directly to the landlord.

Vouchers are not an entitlement, which makes calculating a traditional take-up rate difficult. Only one in five low-income households with unmet housing needs get any form of housing assistance (Kingsley, 2017). In this paper, I focus only on eligible households who are issued a voucher, a small subset of all eligible households. The share of households who are able to use their vouchers once they are issued a voucher is referred to as the success rate; a search is generally considered successful if the household is able to use the voucher within one year of issuance.[2] It's important to note that at this point in the process, households have repeatedly expressed interest in the program and confirmed their eligibility. The voucher program also provides quite generous subsidies. The average household in the sample for this paper received \$8,000 per year, and in more expensive housing markets, this number can be substantially higher. This makes low, and declining, success rates even more striking, and invites further examination of voucher take-up process.

[1] Payment standards are based on HUD's Fair Market Rents (FMRs). PHAs have discretion to set payment standards within 90-110% of FMRs. Voucher holders may rent units above the payment standard, but they are required to pay the difference between the rent of the unit and the payment standard out-of-pocket, and when a voucher recipient joins the program, their rent payment cannot exceed 40% of their income.

[2] The one year window allows for comparable measures across years. Most successful searches will conclude within a year; in the sample for this paper, 99 percent of successful searches ended within a year.

2.2 Frictions in the take-up process

Part of the reason that success rates remain low in the voucher program is that the take-up process requires participants to navigate the private housing market search. In recent years, this has become an increasingly difficult challenge for all low-income households, as rents have risen significantly and a lack of affordable rental homes persists across the country.

Voucher recipients contend with these same challenges, but also have to navigate all the requirements of the program described above, and must do so within a limited time frame. This can create an additional friction for voucher holders, which is that while they may receive their voucher on any day of the month, in general, leases for homes start on the first of the month. This could put voucher recipients who are issued their vouchers toward the end of a month at a disadvantage if they are given, for example, only 60 days to use their voucher. Households who are issued vouchers at the end of month therefore have fewer days of effective search time if they intend to start their lease within 60 days of their voucher being issued, given they are most likely to find leases that start on the first of the month.

If PHAs strictly enforce initial search terms, this could lead to lower success rates for households who start their searches at the end of the month. However, if PHAs offer extensions, or longer initial search terms, we may see only a delay in when households are able to use their vouchers and start receiving benefits, and no change in the overall success rates. Therefore, the ultimate effect of issuance timing can depend on how PHAs set their search time policies, and how they choose to implement certain features of their own policies, such as leniency around issuing extensions.

There also could be potential impacts of issuance timing on the quality of the match between households and the housing units that they find. If households are dealing with a more constrained search window, they may accept housing units that they would otherwise decline, had they been given more time to search. However, some households will take longer to search if their issuance was at the end of the month, because they will roll over and take an additional month to find a housing unit, given that their PHA grants them this flexibility.

3 Data

3.1 Administrative data on program participation

I use detailed administrative data from the Office of Public and Indian Housing (PIH), the division of HUD that is responsible for administering the voucher program. PHAs submit data on each programmatic action to the PIH Information Center (PIC).[3] These actions indicate when each household is issued a voucher, and if and when they are successful in using their voucher. From these actions, I define my key metrics: indicators for a voucher recipient successfully using their voucher, and the elapsed time between the issuance of the voucher and starting a lease with that voucher, for households who are successful. I also create indicators for a household successfully using their voucher within certain key search windows, including 60, 90, 120, 180, and 360 days.

Importantly, I observe key demographics, including race, ethnicity, age, household composition, and origin ZIP Code at the time of voucher issuance. For households who are successful, once they are admitted to the program, I observe their full demographics, complete address, and detailed information on their housing costs and subsidy payments.

For households who are successful in using their vouchers, I also observe all subsequent actions, including moves, ports to other PHAs, income re-certifications, and program exits. From these additional actions, I can construct a history of program participation, and identify how long they remain in the voucher program and at a specific address.[4]

Another key feature of this data is that I observe the ZIP Code where voucher recipients live when they receive their voucher, and where they live once they lease-up, if successful. From this information, I derive a set of neighborhood outcomes that measure whether or not households are able to move to neighborhoods with lower poverty rates. Specifically, I merge the administrative data with data from the 2015-2019 American Community Survey (ACS) on poverty rates and poverty percentiles within a CBSA at the ZCTA level. I identify a dichotomous indicator for moves where the destination poverty percentile is higher or lower than the origin poverty percentile.

Lastly, I use county rent levels from the 1-year 2019 ACS to stratify the sample median country

[3] More detail on this data source can be found in Appendix A.

[4] Specifically, I detect changes in addresses by geocoding each unique address entry. The geocoding match rate is above 95%. I then use the cleaned addresses to detect any changes in address. This methodology ignores moves within the same building, so a move is defined as a change of street address, not simply a change in unit. Changing units within a building is relatively rare; I estimate that only around 3% of moves are within the same building.

rent quartiles to conduct heterogeneity analyses.

3.2 Collected data on PHA search time policies

There is no centralized source of information on allowable search time policies by PHA. PHAs record these policies in annual Housing Choice Voucher Administrative Plans, which do not receive formal approval from HUD, so records of these plans are not maintained. I collect administrative plans from several sources. First, historical plans for 217 PHAs in my administrative data sample were provided by Zhang and Johnson (2023), who hand-collected plans from PHA websites and outreach between 2016 and 2020. An additional 32 plans were obtained from the Boston Regional HUD office via a Freedom of Information Act request. Finally, additional plans were collected directly from PHA websites; however, plans collected from websites generally cover policies going into effect in 2023 and 2024; the administrative data for this paper goes through 2022 only. An additional 23 plans from the years 2015-2022 were gathered directly from PHA websites.

From the administrative plans, I record data on the initial search term, automatic extensions, discretionary extensions, tolling and expiration practices, and any exceptions that PHAs make in search times. I then classify PHAs by their "effective" allowable search time, which is their initial search time plus time given from any automatic extensions. More information about the data collection process can be found in Appendix B.

3.3 Sample

The sample for the administrative data on success rates and search times is expansive, containing 950 PHAs in total. In the years between 2015 and 2022, these PHAs issued over 1.1 million vouchers. I include only PHAs that meet administrative data quality standards as detailed in Ellen et al. (2023), and explained in more detail in Appendix A. I also restrict the sample to include only PHAs that are predominately located in one county, in order to be able to accurately measure market conditions consistently across PHAs. Combined, these PHAs serve around 47% of the total participants in the program nationally, and in general, and very representative of the program as a whole (Appendix table A1).

Table 1 shows important characteristics of the administrative data sample. Around 45% of new voucher recipients are Black, and 14.3% are Hispanic. Additionally, almost three quarters of households are headed by females, with the majority (53%) of these households containing children.

4 Descriptive facts

4.1 Search times and success rates

Low success rates have long been a key challenge for the voucher program, but until recently there was very little insight into voucher success rates or search times. The administrative data used for this paper is one of the only sources of data that allows for measuring both how many households are able to use their vouchers, and how long this process takes on a representative scale and over time.

Prior to 2021, success rates (defined as the share of households who are able to use their voucher within 1 year of issuance) hovered around 65%-68%, but have since declined sharply. By 2022, only 57% of those issued a voucher were able to lease up and enter the program (figure 1, left panel). At the same time, search times have been sharply increasing. In 2022, it took the median voucher recipient around 77 days to use their voucher, up more than two weeks from two years prior (figure 1, right panel). Both of these metrics can indicate growing search costs for voucher recipients.

Another way to measure success rates and to understand the relationship between growing search times and success rates it to look at success rates across different search windows. For instance, a 60-day success rate is defined as the share of voucher recipients who are successful in using their vouchers within 60 days of issuance. This metric is important, considering that voucher recipients are given a set time period to use their voucher; we can use these search windows to see what share of voucher recipients are able to succeed in a given time frame.

Figure 2 shows success rates at 60-, 90-, 120-, 180-, and 360-day search windows, for the years 2015, 2019, and 2022. First, we see that a relatively low share of voucher recipients are able to use their vouchers within 60 days, the federal minimum amount of search time that PHAs are required to give. Even in 2015, when success rates were higher and search times were shorter, only 36% of voucher recipients succeeded in starting leases using their vouchers within 60 days of voucher issuance.

Additionally, it's important to note that success rates declined at every search window between 2015 and 2022, and we see very striking differences when we look across shorter time intervals. For instance, in 2015, around 66% of voucher holders were able to use their voucher within 6 months of receiving it. In 2022, only 50% of voucher recipients were successful within 6 months. This shows that voucher recipients are facing growing search costs, and significant portion of voucher recipients are now searching for more than 6 months to be able to use their voucher.

There is also wide variation in success rates and search times across the country (Figure 3). Ellen et al. (2024b) explain some of this variation with a combination of individual characteristics of households, characteristics of neighborhoods where searches originate, and market conditions, but a large share of this variation still remains unexplained. Differences in search time are particularly large, which can be due to differences in PHA policies, differences in market conditions, and how PHAs may respond to differences in market conditions.

In fact, market conditions have a particularly strong association with search times. Figure 4 shows median search times by county median rent quartile. Successful searches in the highest rent counties are almost twice as long as in the lowest rent counties in 2021 and 2022. However, this does not equate to higher rent counties experiencing lower success rates. In fact, when we look at success rates over a one year search window, higher rent counties actually have considerably higher success rates than lower rent counties (figure 4, left panel). This runs counter to a common prediction that success rates will be lower where markets are tighter; it can also emphasize the complicated relationship between search times and market conditions, as PHAs may set their search time policies or exercise more flexibility within existing policies as a response to tighter housing markets.

4.2 Allowable search time policies

A question that arises from the stylized facts presented thus far is how much of the variation in search time comes from differences in allowable search time, as opposed to differences across households within a certain set of constraints. The data assembled for this paper contain the first ever source of information on PHA search time policies. This is important for several reasons. First, extending search times has been cited as a potential strategy to boost success rates across the program (HUD thing), and as a potential strategy to mitigate racial disparities in success rates (Ellen et al., 2024b). Second, information on PHA policies can be used to determine how flexibility in search times can affect a wide variety of outcomes for participants.

Across the 217 PHAs with available administrative plans from 2016-2020, the majority of PHAs give the federal minimum 60 day initial search term (table 2). However, PHAs almost universally allow extensions, and many PHAs even give automatic extensions.[5] Combined, around 40% of PHAs have a maximum term limit. Maximum term limits serve as a cap on the length of time that a voucher can be extended; these are most common among PHAs with 60 day initial terms. Additionally, most PHAs allow for "tolling", or suspension of the voucher term when a unit is under inspection, which was made mandatory by HUD in 2015. More information about the prevalence of extensions and other characteristics of PHA search time policies and how these characteristics vary over time can be found in Appendix B.

Importantly, however, initial search times are not always closely related to observed search time. PHAs that have a 60 day initial search term actually tend to have longer searches, on average, than PHAs with 120 day search times (table 3). PHAs with 90 day effective initial search terms have the longest observed search times. This indicates that much of the difference in search time across PHAs is likely due to how lenient PHAs are with granting extensions. In fact, at PHAs with a 60-day initial search term, almost half of successful searches succeed more than 15 days past the initial search time. I also construct a measure of a PHA's "effective initial search term". This measure take the initial search terms, and adds the number of days that would be granted from an automatic extension, if the PHA offers automatic extensions. However, we still see that a sizable share of searches succeed past the effective initial search term as well.

5 Effects of issuance timing on success rates

In this section, I use a series of models to document how issuance timing can lead to delays in households' ability to use their vouchers. The timing of voucher issuance can place an additional constraint on voucher recipients who already have to contend with high participation costs and an increasingly difficult housing search. In appendix tables C1 and C2, I show the search vouchers are issued across the days of the month, but searches most commonly conclude on the first of the month, when households are able to start their leases.

Below, I use the following equation to measure the effect of the day of the month that a search

[5] The difference between automatic and discretionary extensions is that in the case of an automatic extensions, generally households must still request the extension, but their request will not be denied. In the case of a discretionary extension, a PHA may deny the request or require the household to meet certain conditions before granting additional search time.

voucher is issued on the probability of successfully using a voucher within a certain search window. The equation takes the following form:

$$Success_{itd} = \beta \sum_{j=1}^{31} \mathbb{I}(d - j = 0) + \ \gamma X'_{itd} + PHA_{itd} \times Year_{it} + \varepsilon \tag{1}$$

where j is an indicator for each day of the month (1-31), d is the day of issuance for household i in year t. Importantly, I use PHA by Year fixed effects to hold constant allowable search time policies at this stage. I also control for individual characteristics, including race/ethnicity, age, household composition, disability status, homeless status, and citizenship status.

These models are identified based on the fact that the exact timing of voucher issuance is random. Several features of the program's administration support this assumption. First, households tend to wait on waitlists for many years, and since many waitlists are commonly closed, in practice many households may even apply on the same day. This means that voucher recipients have no ability to influence when they receive their voucher. Second, even though PHAs may have different prioritization schedules for issuing vouchers, in general these schedules do not line up with the days of the month. Balance tests found in Appendix C support this, showing that there are no significant differences in observed demographics across days of the month.

First, I am using 30-day success rates to emphasize the mechanical relationship between the day of voucher issuance, and when a household is able to lease up (figure 5). As the month progresses, it's harder for households to start a lease by the first of the next calendar month, which leads to fewer households who are able to lease up within 30 days of their issuance. By the end of the month, this effect is around 5 percentage points, which is a 38% decrease from the beginning of the month (only around 13% of searches are successful within 30 days).

Next, I show coefficients on indicator variables for the day of the month of voucher issuance for 4 different search windows: 60-day success, 90-day success, 120-day success, and an unrestricted measure of success rates. In these charts, we see that the effect of timing within a month persists beyond the relationship with 30-day success rates that we see above. Focusing first on the top left panel, we see that search events that begin at the end of the month are significantly less likely to succeed within 60 days of voucher issuance. At the very end of the month, the effect is around 3 percentage points, which is a significant effect given only around 30 percent of searches succeed within 60 days- so it equates to about a 10 percent decline.

The effect of end of month issuance persists when we look at 90 day success rates (top right panel). Here we see that searches that start towards the end of the month are about 1 to 1.5 percentage points less likely to succeed within 90 days. This is a smaller but still noticeable effect, as around 43% of searches succeed within 90 days.

At a 120 day search window, the effect is much more muted; though coefficients are negative at the end of the month and are much lower than some periods in the middle of the month, we don't see a significant effect compared to the reference period at the very beginning of the month. And finally, looking at the bottom right panel, we see no clear patterns when we look at an unrestricted measure of success rates. In other words, we see that end of month issuance has a significant effect on the chance of using a voucher within a shorter period of time, but has no apparent effect on the probability of the household ever using their voucher, at least on average.

In the models that follow, I summarize the effect of day of issuance using an indicator variable for voucher issuance in the last seven days of the month:

$$Success_{itd} = \beta End\,of\,month_{itd} + \ \gamma X'_{itd} + PHA_{itd} \times Year_{it} + \varepsilon \tag{2}$$

Again, I use PHA by year fixed effects to hold constant allowable search time policies, and I control for individual characteristics.

In Figure 7, I show the effect of end of month issuance at every search interval. There is a negative, significant effects at 60, 90, and 120 days, and a marginal effect at 150 days, before the effects attenuate to 0 at longer time periods. Relative to searches that begin in the first three weeks of the month, searches that begin at the end of the month are around 2 percentage points less likely to succeed within 60 days, 1 percentage point less likely to succeed in 90 days, and .75 percentage points less likely to succeed in 120 days. Even at 150 days, we see a close to significant effect of end of month issuance.

These coefficients reflect both the difficulty of using vouchers, and the mechanical relationship between voucher issuance and norms in the private housing market. The fact that we observe significant effects even beyond the 30- and 60-day windows where we would expect the effects of issuance timing to be most pronounced emphasizes how difficult it can be to complete all the steps of program participation, given a complex interaction with the private rental market. The date of issuance matters for the ability of voucher holders to use their vouchers and start their leases in a certain time frame.

One question is how this effect has evolved over time. On the one hand, as housing markets have gotten tighter and the housing search more difficult, there could be a larger effect of losing additional search days. However, it's also possible that in the high search time environment of the past several years, the 60 day threshold has become less salient, so the effects may be minimal. Figure 8 presents coefficients from three different stratified models for different time periods, again using equation 2.

Consistent with expectations, in earlier years (2015-2017), there is a larger penalty for end of month issuance on 60 day success rates. However, when looking at longer search windows, effect sizes are more consistent over time. In 2021 and 2022, we see a small (about half a percentage point) but significant effect even at the 150 time frame. This suggests that as search times have increased, it's more difficult for households to use their vouchers within 150 days, and an issuance date at the end of the month can therefore make a difference. Past 150 days, there is no significant effect of end of month issuance, though the coefficient for 180-day lease-up is marginally significant in later years, again emphasizing how as searches lengthen, longer search windows become more salient.

6 Consequences for match quality

Given that on average, there is no effect of end of month timing on the probability of using a voucher at all, I now turn to evaluating the effects of end of month issuance on match quality outcomes for participants who are successful in using their vouchers. The expected direction of these outcomes is somewhat ambiguous. If households who receive their issuance at the end of the month are more likely to get an extension, then the extra search time may lead to better match quality. However, if households are more rushed to lease up in a shorter time period, we may see declines in housing match quality.

I use three different sets of match quality outcomes. First, I look at the probability of a voucher recipient moving ZIP Codes, and the difference between the poverty rates of origin and destination

ZIP Codes. A lower rate of moving ZIP Codes could indicate that households are more constrained on search time and are not able to look for units that are further away from where they live. Additionally, households that have less time to search may not be able to make an "opportunity move" to a lower poverty ZIP Codes.

I also look at housing turnover. The length of time that a household remains in a housing unit can be interpreted as a measure of match quality; more frequent moves among households can cause additional search and moving costs both for the households themselves and for PHAs. Specifically, I measure the probability that a household moves housing units within one or two years of entering the program.

Lastly, I look at measures of relative unit size. When households are issued their voucher, they qualify for a certain number of bedrooms based on the size and composition of the household.[6] However, households are allowed to rent units of different sizes, as long as the unit still meets the payment standard requirements for the qualified bedroom size. A household is considered "over-housed" if they rent a unit with more bedrooms than they are qualified for. In general, households are not permitted to be "under-housed", or renting a unit with fewer bedrooms than they qualify for; however PHAs are allowed to make exceptions. I therefore consider being over-housed as a positive outcome for households who were able to lease larger units, whereas households who are under-housed may have had to sacrifice for a smaller housing unit in order to use their voucher.

I estimate the effects of end of month issuance on measures of housing match quality using equation 2, with the dependent variable being each binary indicator of match quality described above. In Table 4, I show that end of month issuance is not associated with either expanded neighborhood choice or declines in housing turnover. The coefficient on moving ZIP Codes is significant and negative, though the effect size is quite small, only half a percentage point; those who receive their vouchers at the end of the month are slightly less likely to move ZIP Codes. This lends support to the idea that many of these households may be more constrained on search time, and are not able to move in the time period given. Though this does translate to a slight decline in the change of moving to a *higher* poverty ZIP Code, this is likely just a reflection of households being less likely to move at all.

[6] PHAs have some flexibility in setting occupancy standards, as long as they are not less generous than federal minimum standards.

There is no significant effect on housing unit turnover in the year or two years after admission. Additionally, in the aggregate, there is no effect on the probability of a household being either over- or under-housed. The lack of effects on housing turnover and relative unit size means that end of month issuance may delay program participation for some households, but does not translate into any measurable effect on households using that extra time to find more suitable housing units, and therefore remaining in those homes for longer.

7 Variation in effects of issuance timing by PHA policies

To address the question of the effects of issuance timing under different PHA search time policies, I combine the administrative data on success rates with the collected data on search time policies. I use only search events in the years that I have collected plans for a given PHA. Specifically, in this section, I explore if end-of-month issuance can cause declines in success rates at PHAs with stricter search time policies.

The following models take the form:

$$Success_{itd} = \beta End\,of\,month_{itd} \times Policy\,Feature + \ \gamma X'_{itd} + PHA_{itd} + Year_{it} + \varepsilon \qquad (3)$$

I use a fully interacted model that interacts an indicator for end-of-month issuance with a mutually exclusive set of PHA policies, to preserve the PHA fixed effects.[7] Here, I primarily focus on the effect of end of month issuance on the chance of a household ever using their voucher, under the hypothesis that a stricter search policy could lead to a decline in unrestricted success rates, given the shorter period of time a household would have to use their voucher before it expires. Again, I control for individual characteristics, and I use year fixed effects to control for any aggregate trends over time, as PHAs are observed in only one year between 2015 and 2020.

[7]PHA policies may be endogenous to success rates; for this reason, I compare issuances at different times of the month within PHAs.

7.1 Effects on successful voucher use by PHA search policies

First, I test the effect of end of month issuance on different initial and effective voucher terms (table 5). The difference between the two terms is that the effective search term adds on the number of days that a household would gain from an automatic extension. For instance, if a PHA has an initial search term of 60 days and an automatic extension of 60 days, their effective initial search term is 120 days.

None of the coefficients are statistically significant. This may be due to the fact that nearly all PHAs allow for extensions, including additional discretionary extensions on top of any automatic extensions that they may offer. The result of this is that the term of the voucher becomes a less salient deadline, so even though households may have fewer days of effective search time to lease-up within their 60 day initial search term, they have the ability to easily exceed their initial term and not risk losing their voucher.

To account for this, I then test the effect of end-of-month issuance by type of extension policy. I group PHAs into three mutually exclusive categories: PHAs that offer no extensions, PHAs that offer only discretionary extensions, and PHAs that offer both automatic and discretionary extensions.[8] Table 6 shows these interaction terms, both for all PHAs and for PHAs that have a 60 day initial search term. End-of-month issuances at PHAs with 60 day search terms that do not offer extensions see unrestricted success rates that are 6 percentage points lower than those who are issued vouchers in the first three weeks of the month; however, this effect is only significant when we consider only PHAs that have a 60 day initial search term. This is a meaningful decline, showing that the ability to receive an extension mitigates the effects of reduced search time. At PHAs with the strictest search time policies, issuance timing can affect the ability of households to use their vouchers. For PHAs that allow for either discretionary or both automatic and discretionary extensions, there is no effect of end-of-month issuance on the probability of the household ever using their voucher.

[8] All PHAs that allow automatic extensions also allow for discretionary extensions.

7.2 Effects on match quality by PHA search policies

One reason for the lack of strong effects in the aggregate is that effects on match quality could vary by PHA search time policies. For instance, PHAs with shorter initial search times or less flexible extension policies may see negative effects on match quality, as households contend with a shorter effective search period. For PHAs with longer initial search times or less flexible extension policies, effects are more ambiguous. We may expect no effect, as a longer time period to search would mitigate the harmful effects of a slightly shorter time period to search. On the other hand, households who know they have a longer period of time to search may aim to start a lease a month later, given their later issuance date, and therefore may look at a broader set of housing options or may be more likely to hold out for a housing unit that is a better match.

In order to understand the effects on housing match quality, I first look at the average effects of end-of-month issuance on observed search time for households are PHAs with different search time policies, since search time is the mechanism through which we would except issuance timing to have an effect on different measures of housing match quality. As in the descriptive statistics, search time is measured as the elapsed time between the issuance of a voucher and the household leasing up into the program, for households who are successful in their search. In Table 7, I use equation 3 to show the average effects of end-of-month issuance on observed search time, by both initial search terms and extension policies.

Focusing first on initial search time, PHAs with 90-day search terms see a large, significant decline in search time. This indicates that households at these PHAs are not rolling over to start their lease one month later, and are instead leasing up earlier and searching for shorter periods of time. PHAs that do not offer extensions see the sharpest declines in search time, and especially PHAs with 60 day initial search terms. This set of PHAs with the strictest search time policies see average search times for vouchers that were issued at the end of the month that are over a week shorter than those issued in the first three weeks. This set of PHAs would likely be most susceptible to seeing declines in housing match quality caused by end-of-month issuance, given that it is translating into significant declines in search time.

In Table 8, I again use equation 3 to test the effects of end-of-month issuance on measures of match quality by PHA initial search term. For PHAs with a 60-day initial term, there are no effects on neighborhood outcomes. There is a small but significant decline in the probability of a household moving within 2 years. We also see that successful households are more likely to be over-housed, and less likely to be under-housed. These marginal, positive effect on match quality may indicate that, while on average there are no effects on search time, a subset of households do leverage their issuance date into an extension or additional search time see modest effects on match quality.

Similarly, PHAs with 120-day initial search terms actually see positive effects, in the form of a reduced probability of the household moving housing units within 1 or 2 years of admission to the program. It's possible that at PHAs that offer a longer search term, those that are issued their voucher at the end of the month may be more inclined to spend an additional month searching, and take more time to find units that better meet their preferences.

For PHAs with 90-day initial terms, most effects are insignificant, with the exception of the probability of a household being over-housed, where we see a negative effect of end-of-month issuance. This is consistent with PHAs with 90 day search terms showing lower search times on average for households who are issued their vouchers at the end of the month. This supports the hypothesis that households that face a more constrained search window may be less likely to find a larger unit.

In Table 9, I test the effect of end-of-month issuance by extension policy, again using mutually exclusive categories of no extensions, discretionary extensions only, or both discretionary extensions. For PHAs that do not allow extensions, there is some evidence that households who are issued their voucher at the end of the month are less able to move to lower poverty neighborhoods. There is a positive, significant effect on the probability that the household moves to a higher poverty neighborhood. And, while insignificant, the coefficient on moving to a lower poverty neighborhood is negative. This is in contrast to PHAs with more generous extension policies, which show no effect on either variable.

Additionally, households at PHAs that do not offer extensions are more likely to be under-housed, again reflecting a more constrained search process. This is in contrast to PHAs that allow discretionary extensions, where we see a decline in the probability of being under-housed. This is a notable effect, especially given that these households are also more likely to land in higher poverty neighborhoods; households are more likely to be over-housed (as opposed to under-housed) in neighborhoods with lower rents where larger units may fall within the payment standards for a smaller qualified bedroom size. This is again reflecting the fact that at stricter PHAs, end-of-month issuance can have more substantial effects on how longer a voucher recipients searches for and therefore their ability to find a unit that meets their needs.

8 Discussion

In sum, I find that while, on average, issuance timing does not influence the ability of households to ever use their vouchers, there are significant declines in the ability to use a voucher within a set number of days. While many PHAs have built in the flexibility in their search time policies to mitigate any lasting negative effects of end-of-month issuance, at PHAs with limited flexibility in voucher term, end-of-month issuance can cause a substantial decline in the ability of households to use their vouchers. There is also suggestive evidence that end-of-month issuance combined with a lack of extensions can limit neighborhood choices for those who are successful in using their voucher, and can lead to voucher recipients leasing up into smaller housing units when they are successful.

This work has direct policy implications. First, descriptive findings emphasize the growing difficulty that voucher holders experience in using their vouchers, and high search costs that even successful households entail. While the relationship between allowable search time and success rates is difficult to disentangle, this paper provides the first evidence that allowable search time policies can have meaningful impacts on certain elements of the housing voucher take-up process and potentially on match quality for successful recipients as well.

An encouraging finding is that many PHAs are already giving participants the flexibility in search times to enable success, even in spite of issuance timing that could put them at a disadvantage. The findings in this paper emphasize that extensions are key in enabling households to navigate both the program take-up process and the housing market search process in order to use their vouchers. When extensions are not permitted, PHAs need to be cognizant of issuance timing, and adopt either issuance strategies or extension policies that can alleviate the effects of end of month issuance.

The second policy implication concerns the overall difficulty that voucher recipients have in using their vouchers. More than two in five voucher recipients must return their vouchers to the housing authority in recent years, even despite the fact that many housing authorities already offer flexible search time policies. Strategies such as passing and enforcing Source of Income discrimination laws, reforming inspection practices, and potentially allowing households to move into housing units before inspections are completed when housing markets are tight are important to boost overall success rates and ensure that other factors such as issuance timing don't interfere with the ability of households to use their vouchers.

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Tables and figures

Table 1: Sample summary statistics

	Observations	Share of observations	Success rate (1 year)	Median search time
Black head of household	464143	0.458	0.645	75
Hispanic head of household	145288	0.143	0.649	62
White head of household	354218	0.349	0.627	48
Other race head of household	50117	0.049	0.610	59
Female head of household	735008	0.725	0.641	64
Presence of kids	496339	0.490	0.648	67
Age 65+	105396	0.104	0.673	45
Disability indicator	321208	0.317	0.643	59
Homeless indicator	124821	0.123	0.644	70
All observations	1183946		0.642	63

Figure 1: Success rates and search times, 2015-2022

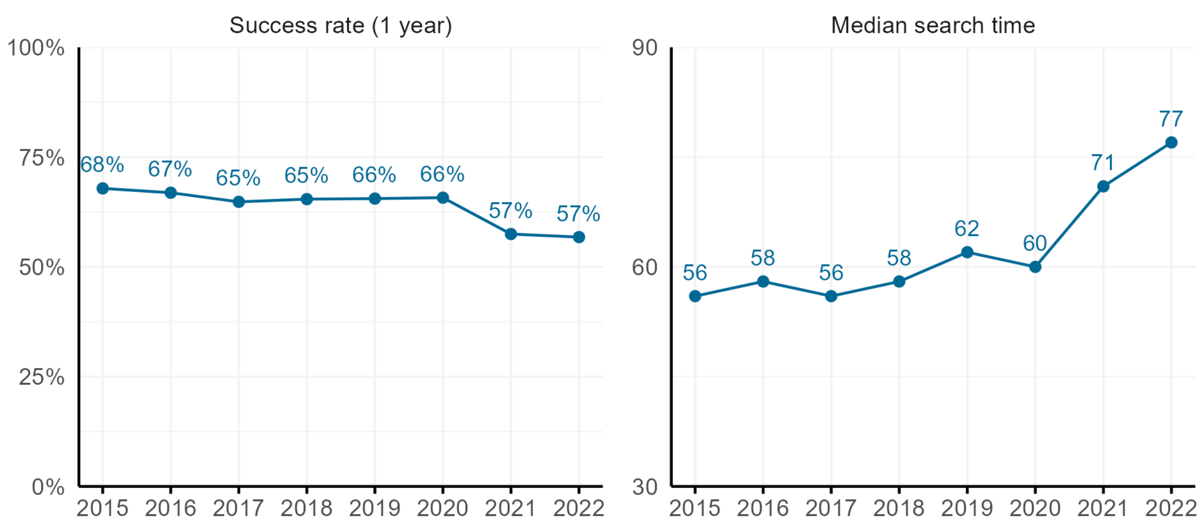


Figure 2: Success rates at different search windows, 2015, 2019, and 2022

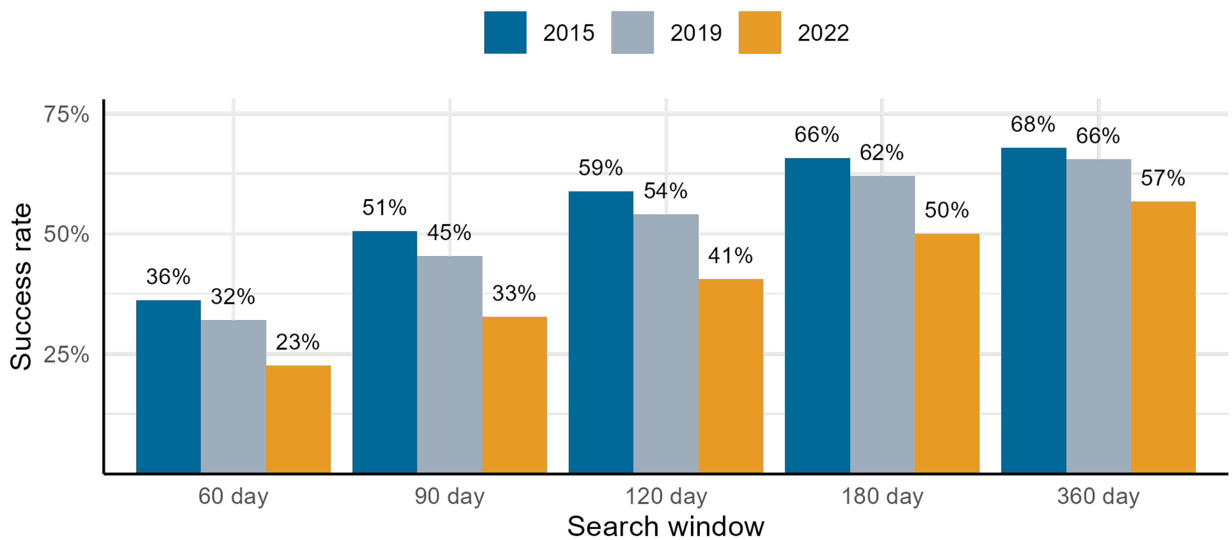


Figure 3: Distribution of success rates and search times, 2021-2022

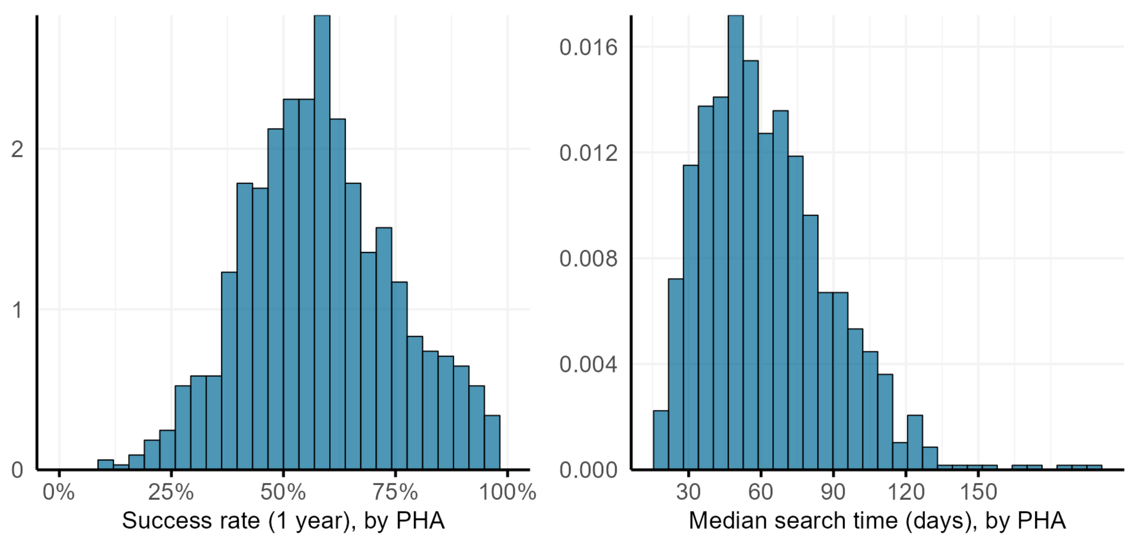


Figure 4: Success rates and search times by county median rent quartile, 2021-2022

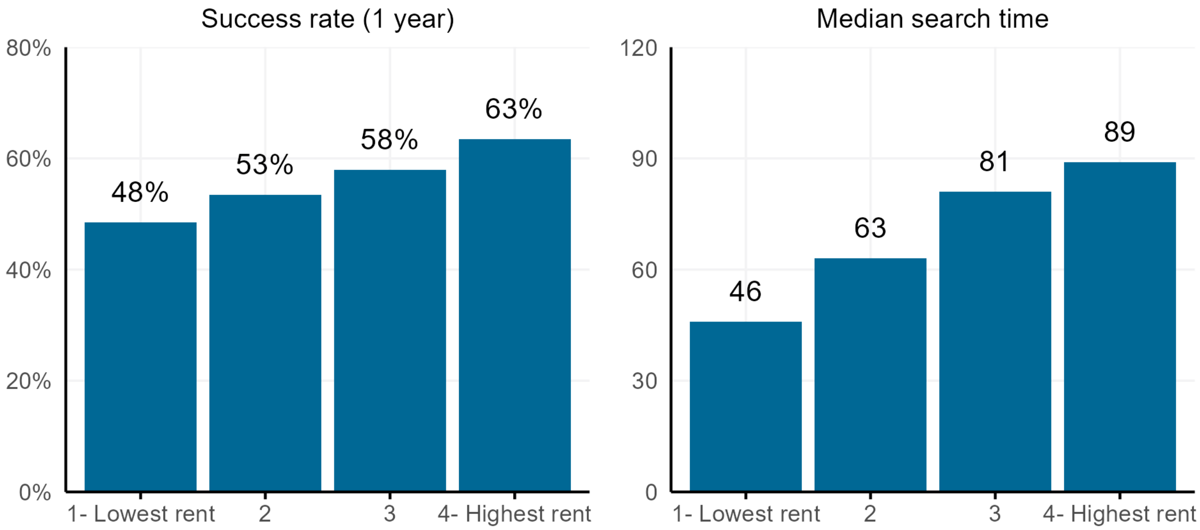


Table 2: PHA search time policy details

Initial voucher term	Number of PHAs	Share of PHAs that do not offer extensions	Share of PHAs that offer discretionary extensions only	Share of PHAs that offer automatic and discretionary extensions	Share of PHAs with a maximum term limit	Share that suspend vouchers during unit insepction
60	168	0.042	0.732	0.226	0.452	0.865
90	16	0.000	0.750	0.250	0.188	0.867
120	24	0.042	0.958	0.000	0.083	1.000

Table 3: Success rates and search time by PHA initial search term policy

Initial voucher term	Number of search events	Success rate (1 year)	Median search time	Share of searches at least 15 days longer than intitial voucher term	Share of searches at lease 15 days longer than effective initial voucher term
60	39419	0.691	72	0.480	0.442
90	6182	0.730	74	0.286	0.271
120	3367	0.738	69	0.171	0.171

Figure 5: Effect of day of issuance on probability of 30-day success

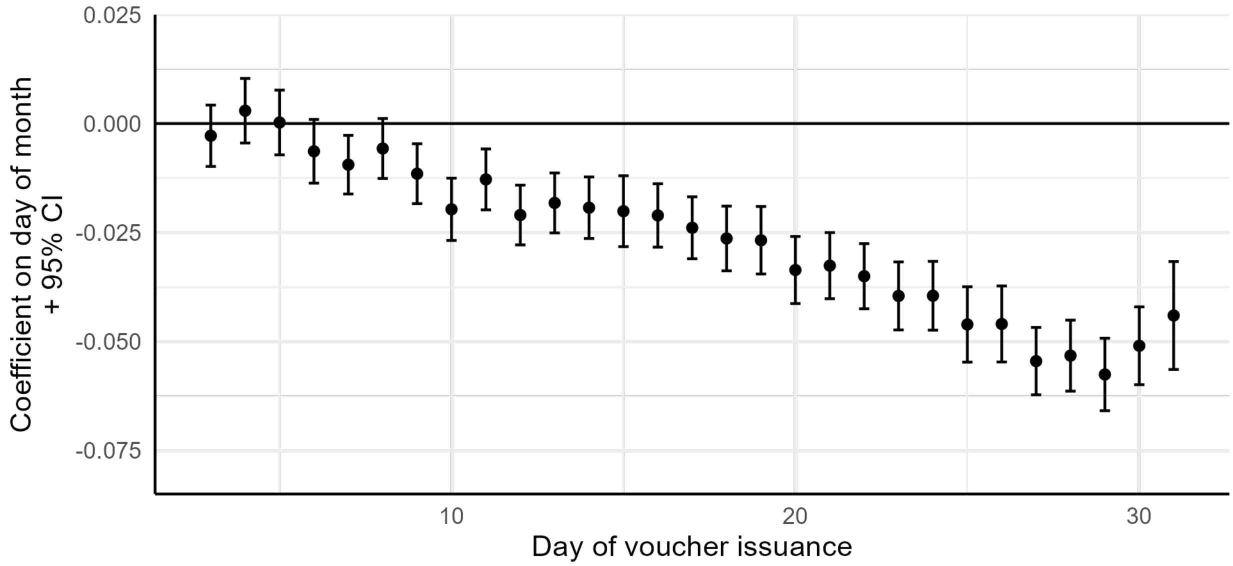


Figure 6: Effect of day of issuance on 60-day, 90-day, 120-day, and unrestricted search window measures of success

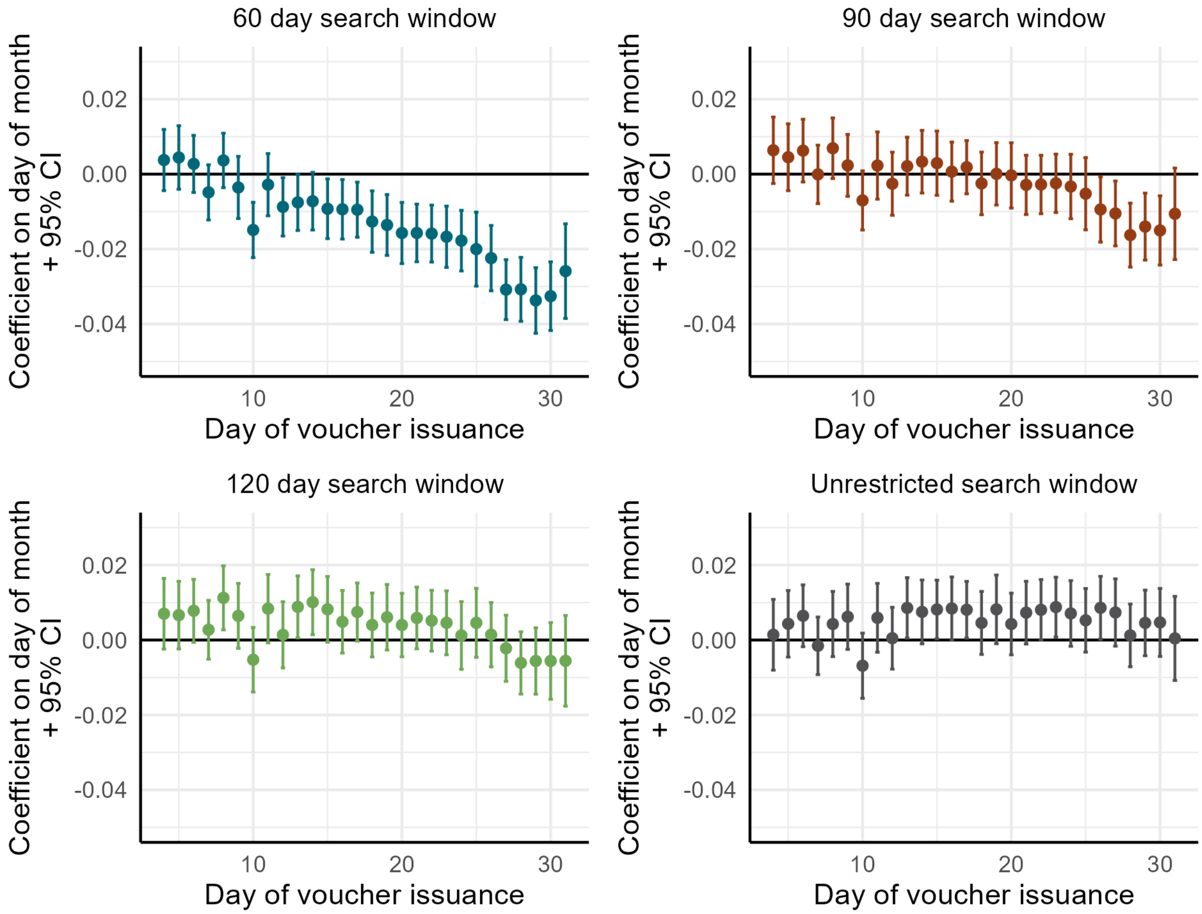


Figure 7: Effect of end-of-month issuance on success at different search windows

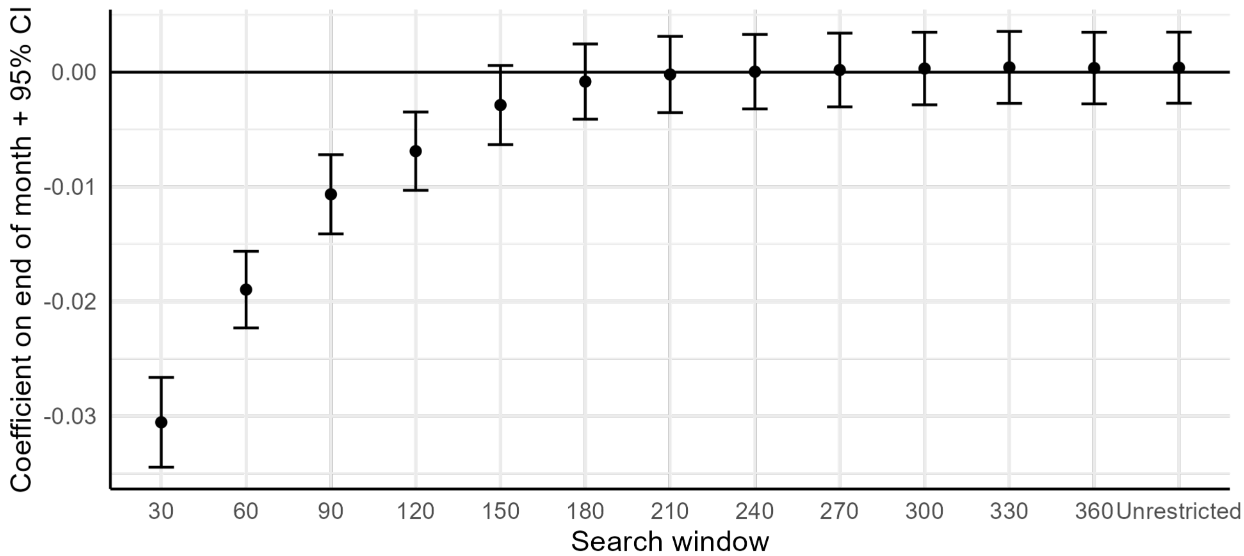


Figure 8: Effect of end-of-month issuance on success at different search windows, over time

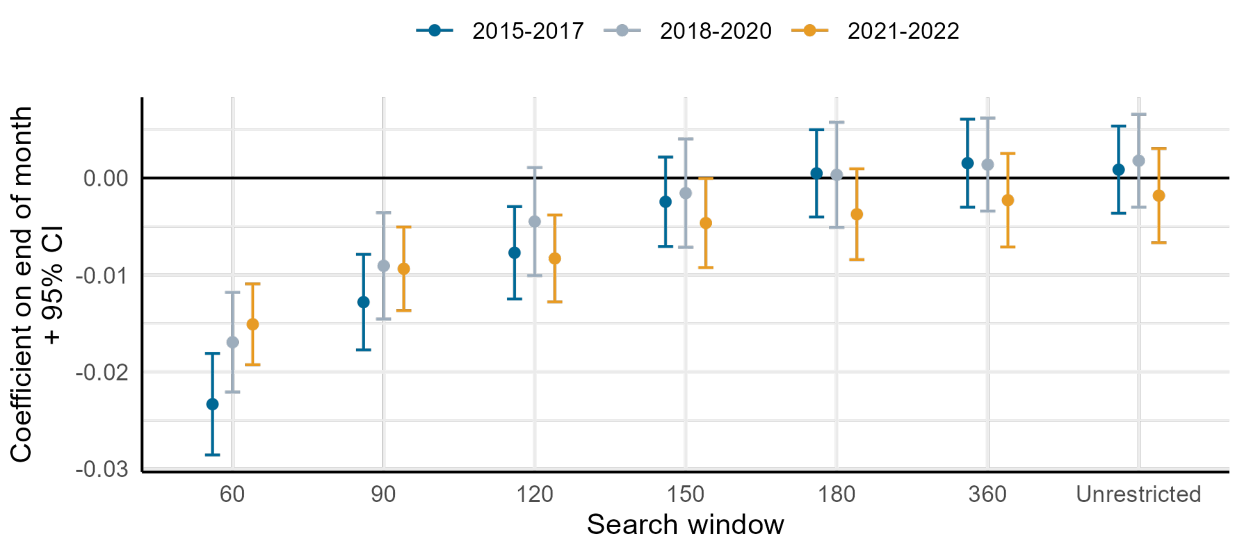


Table 4: Effect of end-of-month issuance on measures of housing match quality

	Move ZIP Codes	Move to lower poverty ZIP Code	Move to a higher poverty ZIP COde	Move within 1 year	Move within 2 years	Over-housed	Under-housed
End of month	-0.004*	-0.001	-0.003*	0.000	-0.001	0.001	0.000
	(0.002)	(0.002)	(0.002)	(0.001)	(0.001)	(0.002)	(0.001)
Num.Obs.	539425	522788	522788	676888	676888	594004	594004
R2	0.203	0.108	0.071	0.025	0.041	0.129	0.053
R2 Adj.	0.192	0.096	0.058	0.014	0.030	0.118	0.041
Standard errors	by PHA	by PHA	by PHA	by PHA	by PHA	by PHA	by PHA
PHA x Year fixed effects	X	X	X	X	X	X	X
Individual controls	X	X	X	X	X	X	X

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 5: Effect of end-of-month issuance on unrestricted measure of success, by PHA initial and effective search terms

	Effect on successful voucher use (unrestricted)	
	Initial search term	Effective search term
End of month x 60 day initial term	0.000	
	(0.007)	
End of month x 90 day initial term	0.026	
	(0.024)	
End of month x 120 day initial term	0.021	
	(0.026)	
End of month x 60 day effective term		-0.002
		(0.010)
End of month x 90 day effective term		0.001
		(0.014)
End of month x 120 day effective term		0.026
		(0.021)
Num.Obs.	41674	41674
R2	0.065	0.065
R2 Adj.	0.061	0.061
Standard errors	by PHA	by PHA
PHA fixed effects	X	X
Year fixed effects	X	X
Individual controls	X	X

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 6: Effect of end-of-month issuance on unrestricted measure of success, by PHA extension policies

	All PHAs	PHAs with 60 day initial terms
	(1)	(2)
End of month x No extensions	-0.029	-0.061*
	(0.025)	(0.023)
End of month x Discretionary extensions only	0.003	0.000
	(0.008)	(0.008)
End of month x Automatic and discretionary extensions	0.013	0.005
	(0.011)	(0.011)
Num.Obs.	43253	35623
R2	0.064	0.070
R2 Adj.	0.060	0.066
Standard errors	by PHA	by PHA
PHA fixed effects	X	X
Year fixed effects	X	X
Individual controls	X	X

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 7: Effect of end-of-month issuance on observed search time, by PHA initial search term and extension policies

	All PHAs		PHAs with 60 day initial term
	(1)	(2)	(3)
End of month x 60 day initial term	-1.312		
	(1.817)		
End of month x 90 day initial term	-6.836***		
	(1.610)		
End of month x 120 day initial term	0.398		
	(1.701)		
End of month x No extensions		-9.277**	-11.487***
		(2.829)	(2.838)
End of month x Discretionary extensions only		-1.357	-2.038
		(1.900)	(2.164)
End of month x Automatic and discretionary extensions		3.430	4.362
		(2.753)	(3.128)
Num.Obs.	28910	30377	24449
R2	0.079	0.078	0.082
R2 Adj.	0.073	0.073	0.077
Standard errors	by PHA	by PHA	by PHA
PHA fixed effects	X	X	X
Year fixed effects	X	X	X
Individual controls	X	X	X

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 8: Effect of end-of-month issuance on measures of housing match quality, by initial search term

	Move ZIP Codes	Move to lower poverty ZIP Code	Move to higher poverty ZIP Code	Move within 1 year	Move within 2 years	Over-housed	Under-housed
End of month x 60 day initial term	-0.003	-0.001	-0.001	-0.001	-0.008+	0.013+	-0.012*
	(0.008)	(0.008)	(0.009)	(0.002)	(0.005)	(0.007)	(0.005)
End of month x 90 day initial term	0.004	0.017	-0.015	0.000	-0.003	-0.029+	0.007
	(0.021)	(0.027)	(0.014)	(0.008)	(0.020)	(0.016)	(0.015)
End of month x 120 day initial term	-0.034	-0.024	-0.010	-0.015**	-0.039*	-0.015	-0.021
	(0.038)	(0.032)	(0.019)	(0.005)	(0.015)	(0.018)	(0.020)
Num.Obs.	24709	24270	24270	28910	28910	28710	28710
R2	0.113	0.064	0.041	0.019	0.032	0.113	0.026
R2 Adj.	0.107	0.058	0.035	0.013	0.027	0.108	0.021
Standard errors	by PHA	by PHA	by PHA	by PHA	by PHA	by PHA	by PHA
PHA fixed effects	X	X	X	X	X	X	X
Year fixed effects	X	X	X	X	X	X	X
Individual controls	X	X	X	X	X	X	X

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 9: Effect of end-of-month issuance on measures of housing match quality, by type of extension policy

	Move ZIP Codes	Move to lower poverty ZIP Code	Move to higher poverty ZIP Code	Move within 1 year	Move within 2 years	Over-housed	Under-housed
End of month x No extensions	-0.005	-0.030	0.020+	0.003	-0.006	0.006	0.013*
	(0.016)	(0.024)	(0.012)	(0.007)	(0.019)	(0.022)	(0.005)
End of month x Discretionary extensions only	-0.009	-0.003	-0.006	-0.001	-0.011*	0.005	-0.012*
	(0.009)	(0.008)	(0.009)	(0.002)	(0.005)	(0.007)	(0.005)
End of month x Automatic and discretionary extensions	0.002	0.009	-0.005	-0.007	-0.007	0.006	-0.015
	(0.020)	(0.018)	(0.025)	(0.007)	(0.009)	(0.015)	(0.011)
Num.Obs.	25718	25254	25254	30377	30377	30172	30172
R2	0.112	0.064	0.040	0.018	0.032	0.116	0.027
R2 Adj.	0.107	0.058	0.034	0.013	0.026	0.111	0.021
Standard errors	by PHA	by PHA	by PHA	by PHA	by PHA	by PHA	by PHA
PHA fixed effects	X	X	X	X	X	X	X
Year fixed effects	X	X	X	X	X	X	X
Individual controls	X	X	X	X	X	X	X

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Appendix

Appendix A: Administrative data on voucher use: data quality standards

This paper uses administrative data from HUD to derive outcomes on success rates and search time. The process to clean these data was extensive; it is described in detail in Ellen et al. (2023). In order to ensure accuracy, I employ data quality standards to ensure that only PHAs with accurate and complete issuance data are included in the sample. In general, data quality standards ensure that I exclude PHAs that do not accurately report their issuance data; these PHAs are likely to have success rates that are biased upwards, as they could be under-reporting failed searches.

There are 1,213 PHAs that meet data quality standards across the seven year panel period from 2015-2022. These PHAs collectively serve around 71% of voucher holders. I then further restrict the sample to PHAs that are serve voucher holders that live predominately in one county, in order to get an accurate measure of housing market conditions; 1,810 PHAs meet this criteria, or 73% of the 2,202 PHAs that served tenant-based voucher holders during this time period. Initial Moving-To-Work (MTW) PHAs are also excluded; these PHAs have different reporting requirements, and do not report origin ZIP Code, which is used to derive some of the key outcomes.

Combining these restrictions results in a final sample of 950 PHAs, which collectively cover 47% of voucher holders (using the population of voucher holders in 2019). Table A1 shows that searches that originate at these PHAs look relatively similar to all PHAs, and all PHAs that meet data quality standards.

Table A1: Balance test for inclusion in sample

Variable	All PHAs	PHAs that meet data quality standards	PHAs predominately covering one county	PHAs in final sample
Black head of household	0.426	0.444	0.445	0.452
Hispanic head of household	0.145	0.140	0.145	0.146
White head of household	0.384	0.371	0.361	0.355
Female head of household	0.726	0.724	0.721	0.723
Presence of children	0.477	0.476	0.476	0.483
Age 65+	0.113	0.110	0.114	0.111
Disabled head of household	0.320	0.322	0.319	0.319
Noncitizen	0.026	0.025	0.027	0.026
Homeless indicator	0.105	0.106	0.108	0.114
Observations	2,207,503	1,712,372	1,608,844	1,036,534

Appendix B: Collecting data on allowable search times

While PHAs are mandated to give voucher recipients 60 days to search when they receive a voucher, there is no maximum amount of time, and PHAs have broad discretion to set initial search time and extension policies. There is, however, no centralized source of information about these policies. From HUD's Housing Choice Voucher guidebook:

> PHAs must give all voucher holders a minimum of 60 days search time. There is, however, no limit on how much longer a PHA may allow a family to search for housing. There are differing opinions on what length of search time is effective in promoting family success in leasing.

HUD also notes that PHAs that use longer search times should maintain data on searches so that they can monitor if extended search times could improve success rates, and to have knowledge of their potential financial liabilities. To my knowledge, this is the first paper that provides descriptive evidence on search time policies.

Each year, PHAs submit administrative plans, where they document a comprehensive set of policies that guide the day-to-day operations of the programs. PHAs are required to document their initial search time, or how many days the voucher is issued for; their policies around extensions, which can include automatic and discretionary extensions; their policies on voucher suspension; and their policies around expiration, or what happens when a household fails to use their voucher in the allotted time given.

For this paper, I collect data on these policies by collecting administrative plans from several sources. First, Zhang and Johnson (2023) provided over 1,200 administrative plans and other documents from PHAs, collected between 2016 and 2020. To my knowledge, this is one of the only sources of historical plans, as most PHAs post only their more recent administrative plans on their websites. Of the documents provided, 217 PHAs are in my empirical sample, and meet data quality standards for the time period. For these PHAs, I can observe their search time policies, and see how these policies are reflected for both failed and successful searches.

Supplementary plans were collected directly from PHA websites, and from a FOIA request to the HUD Boston Regional office. Combined, these sources added an additional 55 administrative plans to the sample obtained from Zhang and Johnson (2023).

In Figure B1 below, I show the distribution of the effective year of the analyzed plans. In the analysis that follow, I use only planes that were in effect prior to 2021, and I exclude plans with an unknown year. In Figure B2, I show that most PHAs tend to adopt the minimum 60 day initial search window. While I do not have many PHAs that are represented across multiple years of the panel, it does appear that the share of PHAs with a 60-day initial search term did decrease in 2019 and 2020, which would be consistent with increasing search times.

Discretionary extensions are almost universal, and many PHAs offer automatic extensions as well (figure B1). Almost all PHAs will suspend the voucher term (known as "tolling") while the PHA completes an inspection; this was made mandatory by HUD in 2015. Additionally, almost all

Figure B1: Effective year of plans analyzed

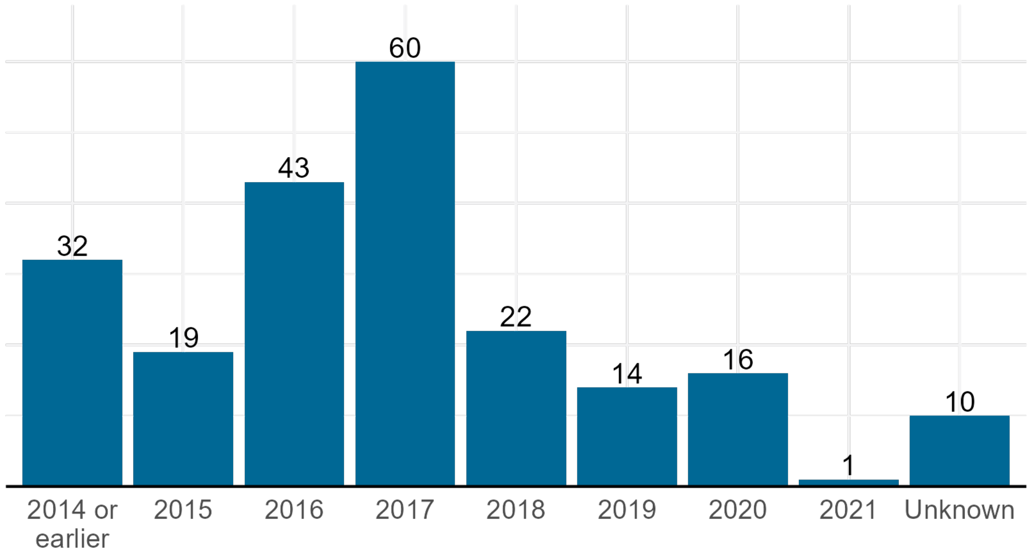
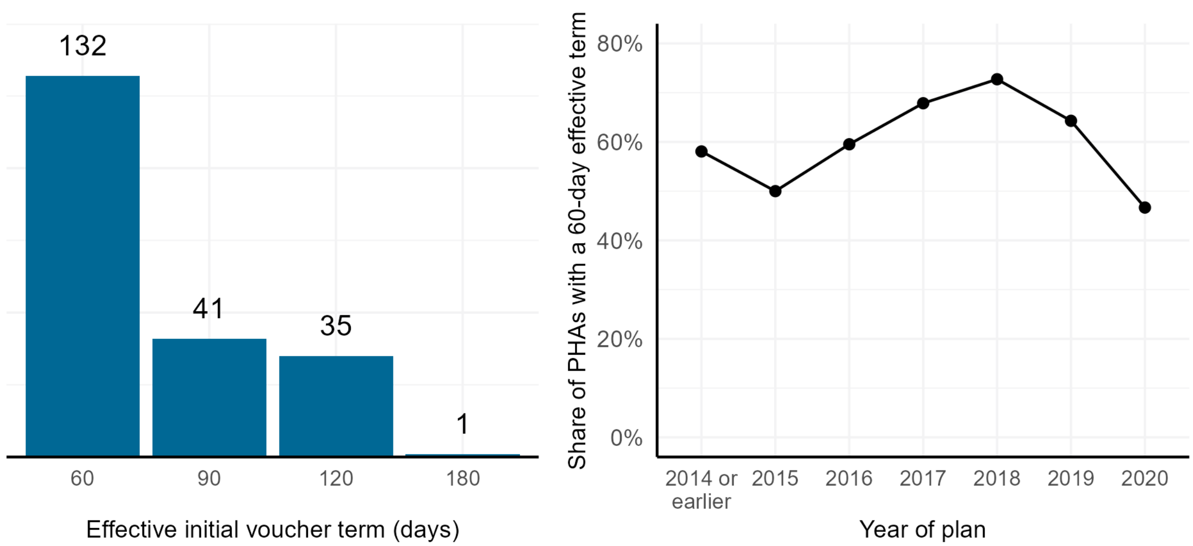


Figure B2: Effective initial voucher term



PHAs over additional time as a reasonable accommodation to voucher recipients with disabilities. PHAs will also often provide extra search time at issuance to households receiving certain types of special purpose vouchers.

Table B1: Share of PHAs that allow for automatic extensions, discretionary extensions, and tolling

Year of plan	Number of PHAs	Share that allow automatic extensions	Share that allow discretionary extensions	Share that suspend voucher term during inspections
2014 or earlier	32	0.156	0.875	0.800
2015	19	0.368	0.895	0.765
2016	43	0.256	0.977	0.923
2017	60	0.117	0.967	0.912
2018	22	0.227	0.955	0.850
2019	14	0.214	1.000	0.929
2020	16	0.375	1.000	1.000

Appendix C: Voucher issuance and lease-up across days of the month

Figure C1: Number of voucher issuance by day of month, 2015-2022

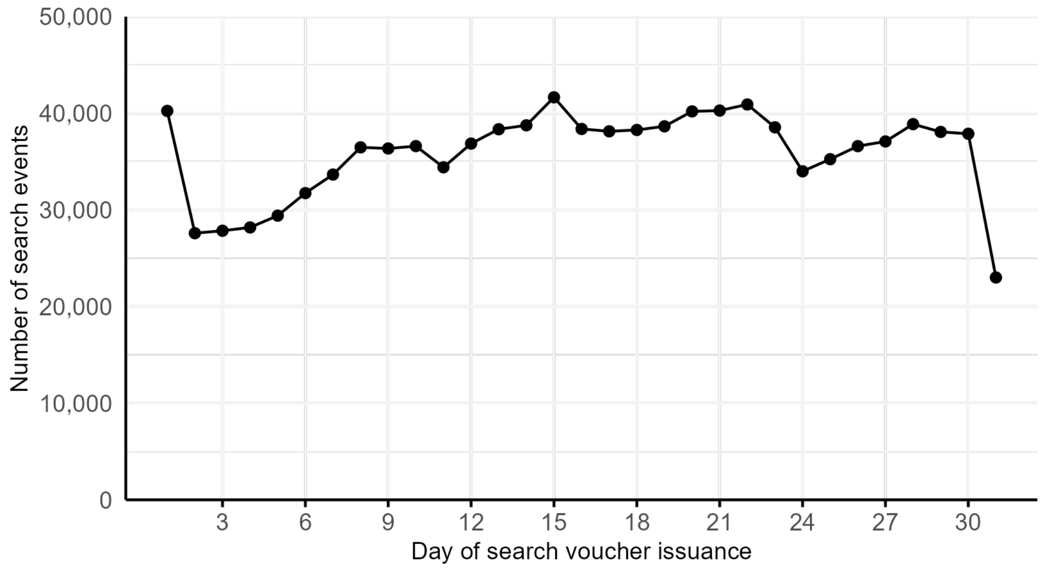


Figure C2: Share of successful searches that lease up on the first of the month, by day of voucher issuance, 2015-2022

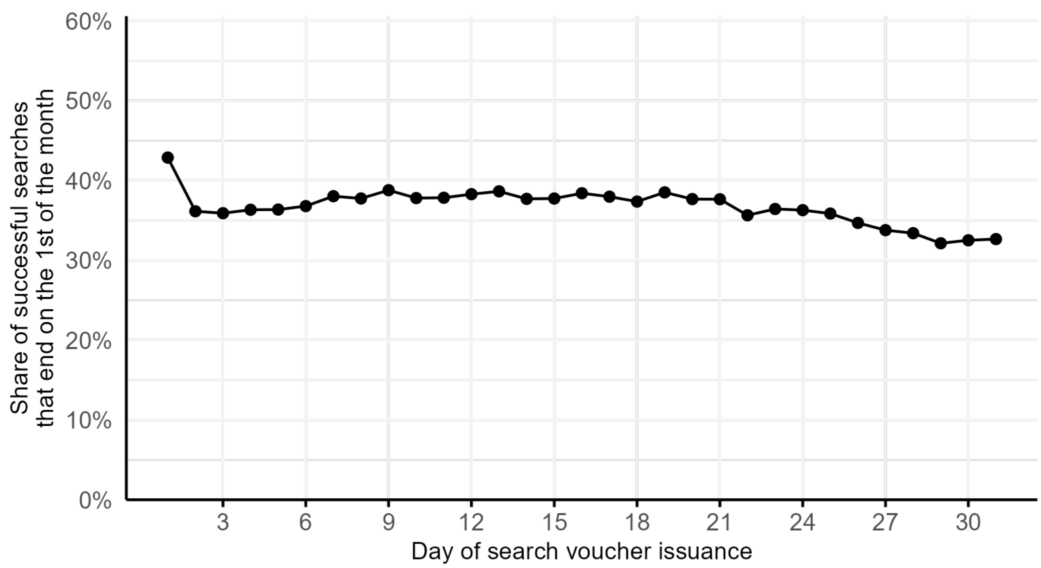


Figure C3: Balance test of individual demographic variables by day of month, 2015-2022